control__eq

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$3 = 3$$

3=3

REFERENCE RENDERING

$$3 = 3$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3 = 3$$

3=3

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__factorial

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$3! = 6$$

```
3!=6
```

REFERENCE RENDERING

$$3! = 6$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
3!=6
```

$$3! = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__indexed_factorial

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$|S_3| = 3! = 6$$

```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__general_factorial

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$|S_n| = n!$$

```
|S_n|=n!
```

REFERENCE RENDERING

$$|S_n| = n!$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{n}|=n!
```

$$|S_n| = n!$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__frac

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



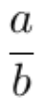
REFERENCE RENDERING



```
\frac{a}{b}
```

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

PP-FormulaNet-L

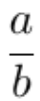
Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None



```
\frac{a}{b}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__frac_sum

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\frac{a+b}{c-d}$$

```
\frac{a+b}{c-d}
```

REFERENCE RENDERING

$$\frac{a+b}{c-d}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
\frac{a+b}{c-d}
```

$$\frac{a+b}{c-d}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__nested_frac

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

REFERENCE RENDERING

$$\frac{1}{1+\frac{1}{x}}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+\frac{1}{x}}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$\frac{1}{1+\frac{1}{x}}$$

```
\frac{1}{1+{\frac{1}{x}}}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__long_fraction

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}
```

REFERENCE RENDERING

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}
```

$$y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}
```

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}
```

$$y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}
```

$$y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__vec2

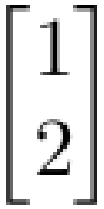
control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1\\2\end{bmatrix}$$

```
\begin{bmatrix}1\\2\end{bmatrix}
```

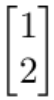
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

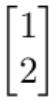
```
\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

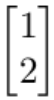
```
\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{1}{2}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__vec3

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1\\2\\3\end{bmatrix}$$

```
\begin{bmatrix}1\\2\\3\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1\\2\\3\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{1}{2}
```

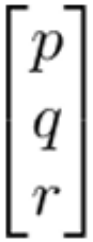
$$\binom{1}{2}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__vec_symbolic

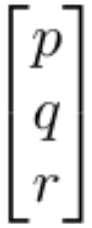
control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

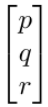
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

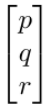
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

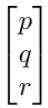
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

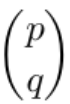
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{p}{q}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__matrix22

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$$

```
\begin{bmatrix}1&2\\3&4\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}\\ {3}&{4}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\begin{array}{r}{\left[1\quad2\right]}\\ {3\quad4\big]}\end{array}
```

$$\begin{array}{r}{\left[1\quad2\right]}\\ {3\quad4\big]}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__matrix23

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\begin{array}{r}{\left[1\quad2\quad3\right]}\\ {4\quad5\quad6\big]}\end{array}
```

$$\begin{array}{r}{\left[1\quad2\quad3\right]}\\ {4\quad5\quad6\big]}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__matrix33

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{array}{l l l}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\end{array}\right]
```

$$\left[\begin{array}{l l l}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\end{array}\right]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__augmented

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]$$

```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\left[\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\ }\end{array}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{array}{c c||c}{1}&{-3}&{0}\\ {2}&{4}&{0}\\ \end{array}\right]$$

```
\left[\begin{array}{c c||c}{1}&{-3}&{0}\\ {2}&{4}&{0}\\ \end{array}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\end{array}\right]$$

```
\left[{\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\end{array}}\right]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$\left[\begin{array}{l l|l}{1}&{-3}\\ {2}&{~4}\end{array}\middle|\begin{array}{l}{0}\\ {0}\end{array}\right]$$

```
\left[\begin{array}{l l|l}{1}&{-3}\\ {2}&{~4}\end{array}\middle|\begin{array}{l}{0}\\ {0}\end{array}\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__line1

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$3x = 1$$

```
3x=1
```

REFERENCE RENDERING

$$3x = 1$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__lines2

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}$$

```
\begin{aligned}3x&=1\\x+2y&=3\end{aligned}
```

REFERENCE RENDERING

$$\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{{x}=1,}\\ {{x}+2{y}=3.}\\ \end{array}
```

$$\begin{array}{r}{{x}=1,}\\ {{x}+2{y}=3.}\\ \end{array}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__lines3

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{aligned}x+y&=2\\x-y&=0\\z&=3\end{aligned}$$

```
\begin{aligned}x+y&=2\\x-y&=0\\z&=3\end{aligned}
```

REFERENCE RENDERING

$$\begin{aligned}x+y&=2\\x-y&=0\\z&=3\end{aligned}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}
```

$$\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{aligned}{x+y}&{{}=2}\\ {x-y}&{{}=0}\\ {z}&{{}=3}\\ \end{aligned}
```

$$\begin{aligned}{x+y}&{{}=2}\\ {x-y}&{{}=0}\\ {z}&{{}=3}\\ \end{aligned}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}
```

$$\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}
```

$$\begin{array}{r}{x+y=2}\\ {x-y=0}\\ {z=3}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__sup

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



```
x^2
```

REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
x^{2}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__sub

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING



```
x_i
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None



```
x_{i}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__sub_sup

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$x_i^2 + y_j^3$$

```
x_i^2+y_j^3
```

REFERENCE RENDERING

$$x_i^2 + y_j^3$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

$$x_i^2 + y_j^3$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

$$x_i^2 + y_j^3$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

$$x_i^2 + y_j^3$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
x_{i}^{2}+y_{j}^{3}
```

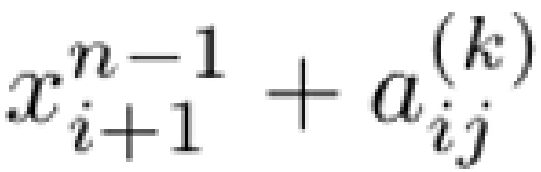
$$x_i^2 + y_j^3$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__complex_scripts

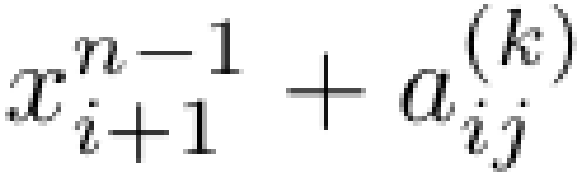
control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT



```
x_{i+1}^{n-1}+a_{ij}^{(k)}
```

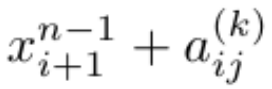
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

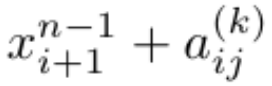
```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

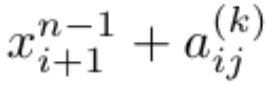
```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

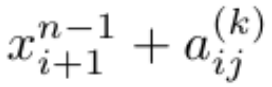
```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
x_{i+1}^{n-1}+a_{i j}^{(k)}
```



Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__set

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$S = \{1, 2, 3\}$$

```
S=\{1,2,3\}
```

REFERENCE RENDERING

$$S = \{1, 2, 3\}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
S=\{1,2,3\}
```

$$S = \{1, 2, 3\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

control__set_text

control | native_tex_180dpi | no GT-based routing | all models, same input

ACTUAL INPUT

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\{A_{3\times3}\mid A\text{ is a permutation matrix}\}
```

REFERENCE RENDERING

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A \mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}\}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{mathrm s a a p e r m u t a t i o n m a t r i x}\}
```

$$S_3 = \{A_{3\times3} \mid A mathrm s a a p e r m u t a t i o n m a t r i x\}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A \mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}\}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

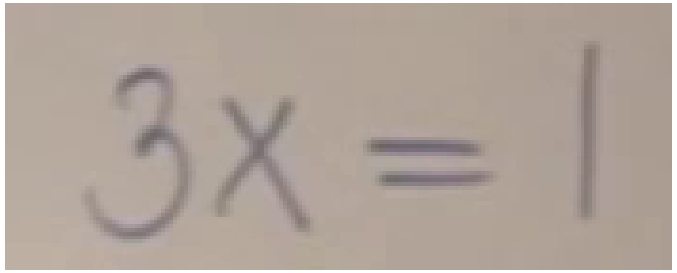
$$S_3 = \{A_{3\times3} \mid A \mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_01_10423__rgb__formula

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
3x=1
```

REFERENCE RENDERING

$$3x = 1$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

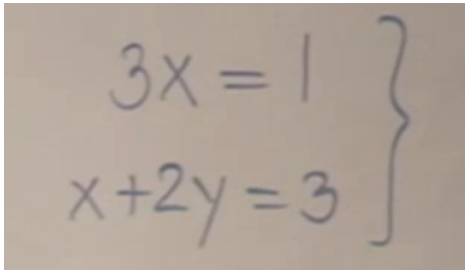
```
3x=1
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_01_10423__rgb__formula_system

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT




```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3+2y=33$$

```
3+2y=33
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$x+2y=3$$

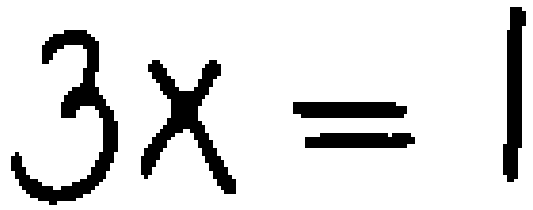
```
x+2y=3
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_01_10423__b0__formula

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
3x=1
```

REFERENCE RENDERING

$$3x = 1$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
3x=1
```

$$3x = 1$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
3x=1
```

$$3x = 1$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
3x=1
```

$$3x = 1$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
3x=1
```

$$3x = 1$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_01_10423__b0__formula

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

```
3x=1
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

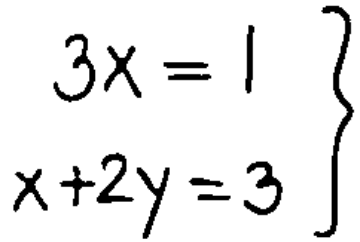
```
3x=1
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_01_10423__b0__formula_system

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: failed | Matrix cells: None

```
3left+x=1\right\
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=3$$

```
3x=3
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$\{3x=1\}\{x+2y=3\}$$

```
\{3x=1\}\{x+2y=3\}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$x+2y=3$$

```
x+2y=3
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_01_10423__b0__formula_system

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$\left.\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}\right\}$$

```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\left.\begin{array}{c}{{3x=1}}\\ {{x+2y=3}}\end{array}\right\}
```

$$\left.\begin{array}{c}{{3x=1}}\\ {{x+2y=3}}\end{array}\right\}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\left.\begin{array}{r}{3x=1}\\ {x+2y=3}\\ \end{array}\right\}
```

$$\left.\begin{array}{r}{3x=1}\\ {x+2y=3}\\ \end{array}\right\}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\left.\begin{aligned}{3x}&{{}=1}\\ {x+2y}&{{}=3}\end{aligned}\right\}
```

$$\left.\begin{aligned}{3x}&{{}=1}\\ {x+2y}&{{}=3}\end{aligned}\right\}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1\bigg\}}\\ {x+2y=3\bigg\}}\end{array}
```

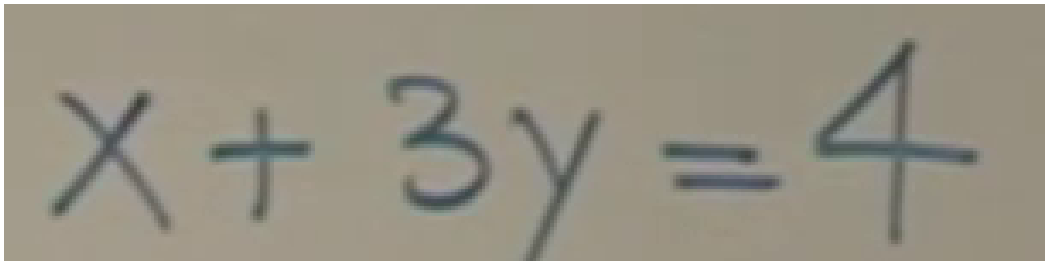
$$\begin{array}{r}{3x=1\bigg\}}\\ {x+2y=3\bigg\}}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_02_44601__rgb__formula

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



x+3y=4

REFERENCE RENDERING

$$x + 3y = 4$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

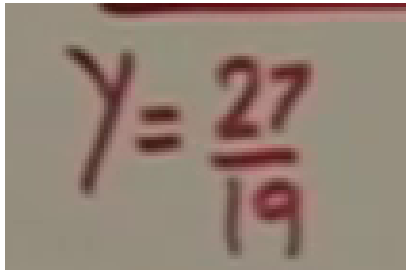
x+3y=4

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_02_44601__rgb__formula_fraction

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
y=\frac{27}{19}
```

REFERENCE RENDERING

$$y = \frac{27}{19}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_02_44601__b0__formula

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



x+3y=4

REFERENCE RENDERING

$$x + 3y = 4$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 3y = 4$$

x+3y=4

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_02_44601__b0__formula

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



x+3y=4

REFERENCE RENDERING

$$x + 3y = 4$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

x+3y=4

$$x + 3y = 4$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

x+3y=4

$$x + 3y = 4$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

x+3y=4

$$x + 3y = 4$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

x+3y=4

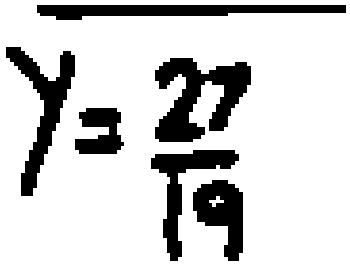
$$x + 3y = 4$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_02_44601__b0__formula_fraction

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
y=\frac{27}{19}
```

REFERENCE RENDERING

$$y = \frac{27}{19}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y=\frac{2pi}{sqrt{g}}
```

$$y = \frac{2pi}{sqrtg}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

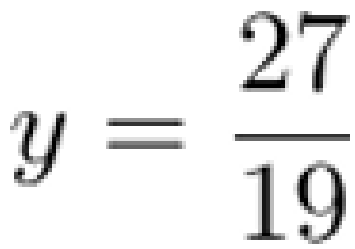
$$y = \frac{27}{19}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_02_44601__b0__formula_fraction

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



```
y=\frac{27}{19}
```

REFERENCE RENDERING

$$y = \frac{27}{19}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{27}{19}}
```

$$y = \frac{27}{19}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

$$y = \frac{27}{19}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{27}{19}}
```

$$y = \frac{27}{19}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{27}{19}
```

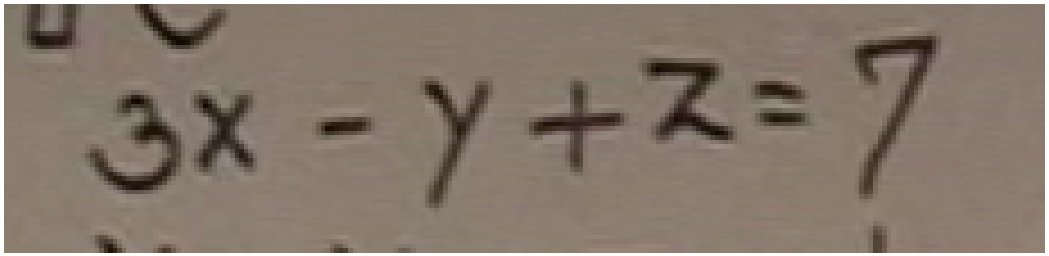
$$y = \frac{27}{19}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__nm_lecture_01_19899__rgb__formula

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT




```
3x-y+z=7
```

REFERENCE RENDERING

$$3x - y + z = 7$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

```
3x-y+z=7
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

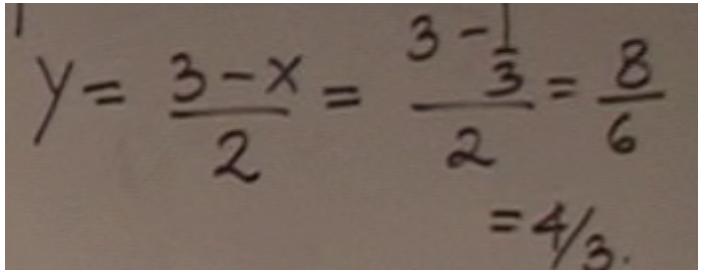
```
3x-y+z=7
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__nm_lecture_01_19899__rgb__formula_fraction

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT




```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

REFERENCE RENDERING

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}=\frac{8}{6}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6} = \frac{8}{6}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

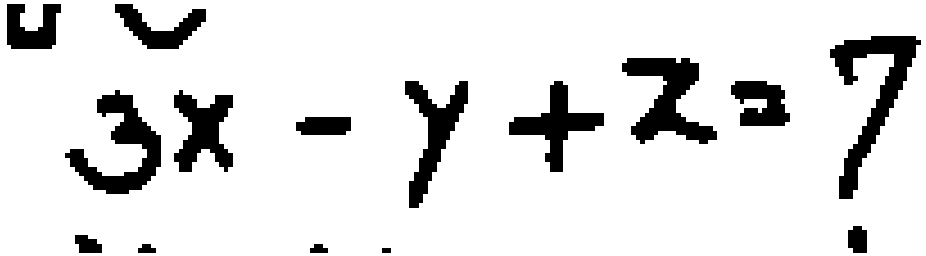
$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__nm_lecture_01_19899__b0__formula

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
3x-y+z=7
```

REFERENCE RENDERING

$$3x - y + z = 7$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

```
3x-y+z=7
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x - y + x = 7$$

```
3x-y+x=7
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

```
3x-y+z=7
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x - y + z = 7$$

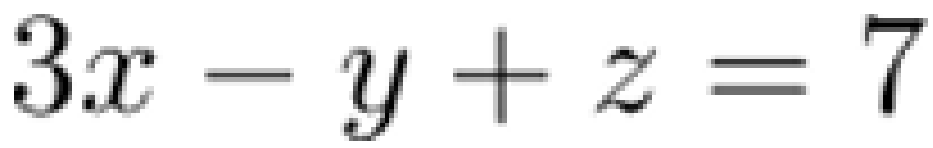
```
3x-y+z=7
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__nm_lecture_01_19899__b0__formula

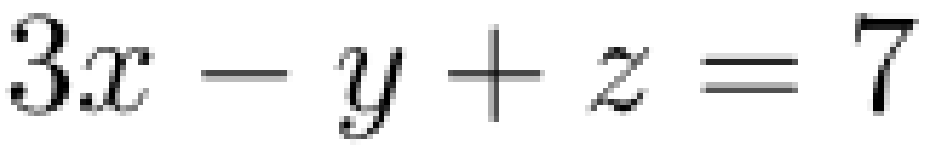
matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



```
3x-y+z=7
```

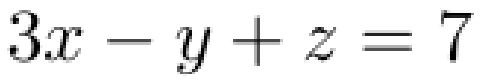
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

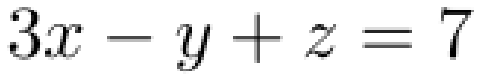
```
3x-y+z=7
```



PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

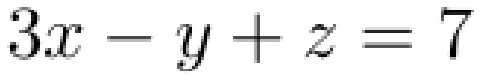
```
3x-y+z=7
```



PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

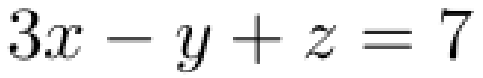
```
3x-y+z=7
```



UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
3x-y+z=7
```

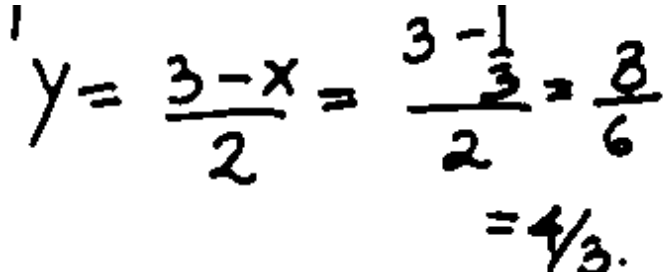


Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__nm_lecture_01_19899__b0__formula_fraction

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

REFERENCE RENDERING

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{3}{6}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{3}{6}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}=4_{3.}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6} = 4_{3.}$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

$$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__nm_lecture_01_19899__b0__formula_fraction

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

REFERENCE RENDERING

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}={\frac{8}{6}}
```

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}
```

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}={\frac{8}{6}}
```

$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
y={\frac{3-x}{2}}={\frac{3-{\frac{1}{3}}}{2}}={\frac{8}{6}}
```

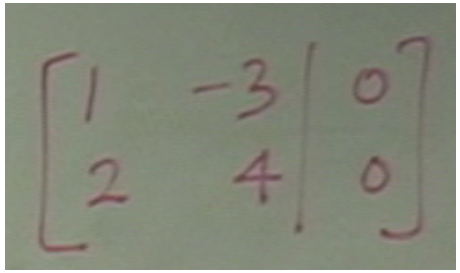
$$y=\frac{3-x}{2}=\frac{3-\frac{1}{3}}{2}=\frac{8}{6}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__rgb__augmented_matrix

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2-310]
```

$$[2-310]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2-3]07
```

$$[2-3]07$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[1-3\vert0]
```

$$[1-3|0]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[1,-3,0]
```

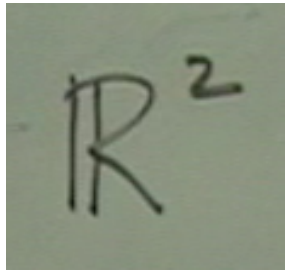
$$[1,-3,0]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__rgb__real_space

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\mathbb{R}^{2}
```

REFERENCE RENDERING

$$\mathbb{R}^2$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
R{}^{2}
```

$$R^2$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
R^{2}
```

$$R^2$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
R^{2}
```

$$R^2$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
R^{2}
```

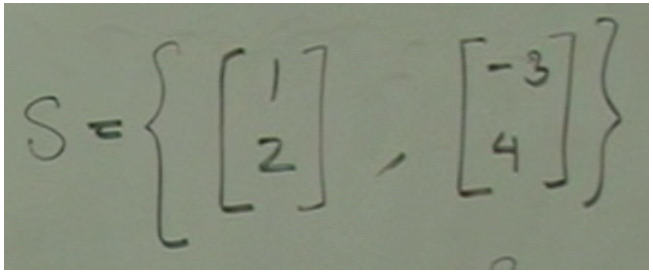
$$R^2$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__rgb__vector_set

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
S=\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}-3\\4\end{bmatrix}\right\}
```

REFERENCE RENDERING

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
S=\{[2],[3]\}
```

$$S = \{[2], [3]\}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
S=\{[2],[3]\}
```

$$S = \{[2], [3]\}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
S=\{[1\frac{1}{2}],[\frac{-3}{4}]\}
```

$$S = \{[1\frac{1}{2}], [\frac{-3}{4}]\}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
S=\{[\frac{1}{2}],[\frac{-3}{4}]\}
```

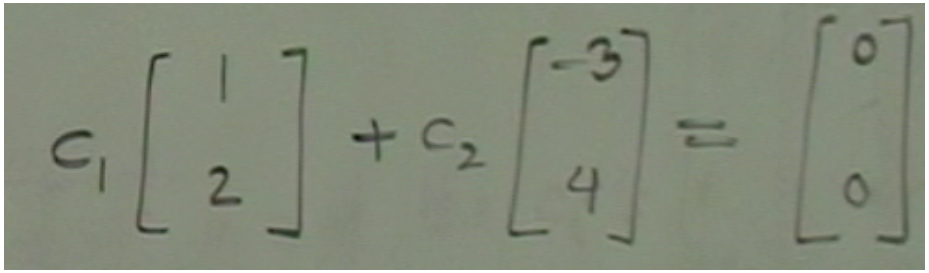
$$S = \{[\frac{1}{2}], [\frac{-3}{4}]\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__rgb__linear_combination

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
c_1\begin{bmatrix}1\\2\end{bmatrix}+c_2\begin{bmatrix}-3\\4\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}
```

REFERENCE RENDERING

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
c_{1}[2]+c_{2}[-3]=[0]
```

$$c_1[2] + c_2[-3] = [0]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
c_{1}[\frac{1}{2}]+c_{2}[\frac{-3}{4}]=[0]
```

$$c_1[\frac{1}{2}] + c_2[\frac{-3}{4}] = [0]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
c_{1}[2]+c_{2}[4]=[0]
```

$$c_1[2] + c_2[4] = [0]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
c_{1}[1]+c_{2}[3]=[0]
```

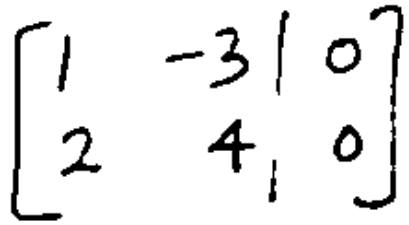
$$c_1[1] + c_2[3] = [0]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__b0__augmented_matrix

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2,310]]
```

$$[2, 310]]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2,\ frac-3]0o]
```

$$[2,\ frac - 3]0o]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[1-310]
```

$$[1 - 310]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{array}{l l l}{1}&{-3\vert}&{0}\\ {2}&{4}&{0}\end{array}]
```

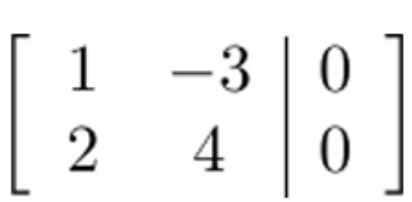
$$[\begin{array}{l l l}{1}&{-3\vert}&{0}\\ {2}&{4}&{0}\end{array}]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

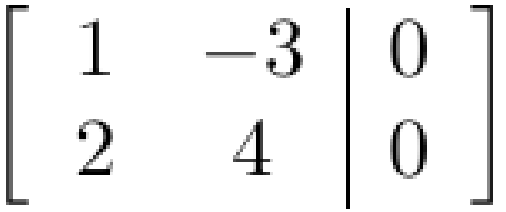
matched__lecture_06_47799__b0__augmented_matrix

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

PP-FormulaNet-S

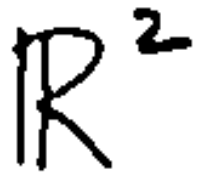
Strict EM: False | Render: unsupported | Matrix cells: False

```
\left[\begin{array}{c c|c}{1}&{-3}&{0}\\ {2}&{4}&{0}\ \ \ \
```

real__lecture_06_47799__b0__real_space

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\mathbb{R}^{2}
```

REFERENCE RENDERING

$$\mathbb{R}^2$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
R^{z}
```

$$R^z$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
P^{z}
```

$$P^z$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
R^{2}
```

$$R^2$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
R^{2}
```

$$R^2$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_06_47799__b0__real_space

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



```
\mathbb{R}^{2}
```

REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

```
\mathbb{R}^{2}
```



PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

```
\mathbb{R}^{2}
```



PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

```
\mathbb{R}^{2}
```



UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

```
\mathbb{R}^{2}
```

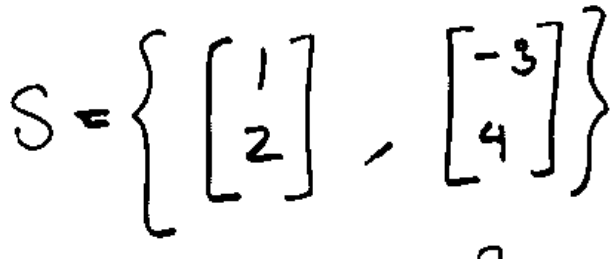


Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__b0__vector_set

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
S=\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}-3\\4\end{bmatrix}\right\}
```

REFERENCE RENDERING

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right\}$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
S=\int\[\frac{1}{\sum2}]_{j}^{-1}]_delta{j}^{-}}\\\\ endend
```

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
s to int limits{{}}}_{{d}}sim{{}
```

PP-FormulaNet-L

Strict EM: False | Render: unsupported | Matrix cells: False

```
S=\{[1\atop2],[4]\}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
S=\{[\frac{1}{2}],[\frac{-3}{4}]\}
```

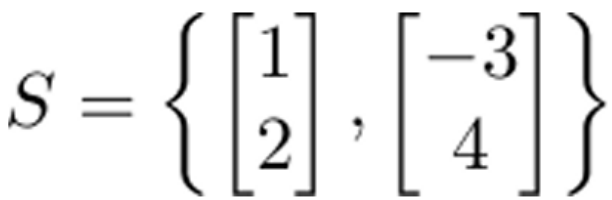
$$S = \{[\frac{1}{2}], [\frac{-3}{4}]\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

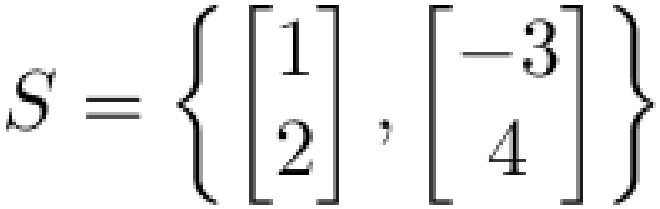
matched__lecture_06_47799__b0__vector_set

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING



```
S=\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}-3\\4\end{bmatrix}\right\}
```

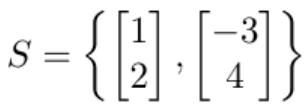
PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
S=\left\{\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right],\left[\begin{matrix}{-3}\\ {4}\ {\right{}}\right{\}}\end{array}
```

PP-FormulaNet_plus-S

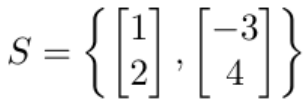
Strict EM: False | Render: rendered | Matrix cells: True



```
S=\left\{\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right],\left[\begin{matrix}{-3}\\ {4}\\ \end{matrix}\right]\right\}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True



```
S=\left\{\left[\begin{matrix}{1}\\ {2}\end{matrix}\right],\left[\begin{matrix}{-3}\\ {4}\end{matrix}\right]\right\}
```

UniMERNet

Strict EM: False | Render: unsupported | Matrix cells: False

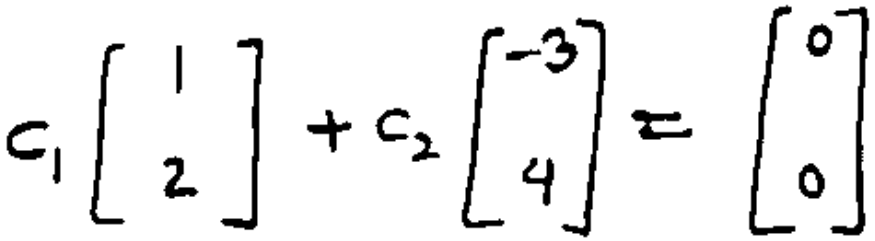
```
S=\left\{{\textstyle\left[1\atop2\right]},{\textstyle\left[{-3\atop4}\right]}\right\}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_06_47799__b0__linear_combination

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

```
c_1\begin{bmatrix}1\\2\end{bmatrix}+c_2\begin{bmatrix}-3\\4\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$c_1[\frac{1}{2}]+c_2[\frac{-1}{2}]=\sum 0limits_0^1$$

```
c_{1}[\frac{1}{2}]+c_{2}[\frac{-1}{2}]=\sum0limits_{0}^{1}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
a_{1}[i\ frac{1}{z2}]+a_{2}[\frac{3}{z}]^{-3}[\]{2}[\frac{1}{z}]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$c_1[2]+c_2[4]=[0]$$

```
c_{1}[2]+c_{2}[4]=[0]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$c_1[2]+c_2[3]=[0]$$

```
c_{1}[2]+c_{2}[3]=[0]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_06_47799__b0__linear_combination

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$c_1 \begin{bmatrix}1\\2\end{bmatrix} + c_2 \begin{bmatrix}-3\\4\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

REFERENCE RENDERING

$$c_1 \begin{bmatrix}1\\2\end{bmatrix} + c_2 \begin{bmatrix}-3\\4\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

```
c_1\begin{bmatrix}1\\2\end{bmatrix}+c_2\begin{bmatrix}-3\\4\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

$$c_1 \begin{bmatrix}1\\2\end{bmatrix} + c_2 \begin{bmatrix}-3\\4\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

```
c_{1}\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]+c_{2}\left[\begin{matrix}{-3}\\ {4}\\ \end{matrix}\right]=\left[\begin{matrix}{0}\\ {0}\\
\end{matrix}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

$$c_1 \begin{bmatrix}1\\2\end{bmatrix} + c_2 \begin{bmatrix}-3\\4\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

```
c_{1}\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]+c_{2}\left[\begin{matrix}{-3}\\ {4}\\ \end{matrix}\right]=\left[\begin{matrix}{0}\\ {0}\\
\end{matrix}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

$$c_1 \begin{bmatrix}1\\2\end{bmatrix} + c_2 \begin{bmatrix}-3\\4\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

```
c_{1}\left[\begin{matrix}{1}\\ {2}\\ \end{matrix}\right]+c_{2}\left[\begin{matrix}{-3}\\ {4}\\ \end{matrix}\right]=\left[\begin{matrix}{0}\\ {0}\\
\end{matrix}\right]
```

UniMERNet

Strict EM: False | Render: unsupported | Matrix cells: False

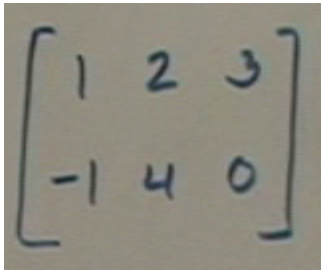
```
c_{1}\left[1\atop2\right]+c_{2}\left[-3\atop4\right]=\left[0\atop0\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__rgb__rectangular_matrix

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}1&2&3\\-1&4&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$[122]]$$

```
[122]]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$[[2]3]$$

```
[[2]3]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[123]$$

```
[123]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$[-140]$$

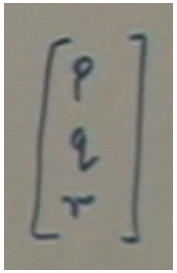
```
[-140]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__rgb__symbolic_vector

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\left[\begin{matrix}{9}\\ {9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{9}\\ {9}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
l
```

$$l$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{9}{9}]
```

$$[\frac{9}{9}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{array}{l}{P}\\ {q}\\ {r}\end{array}]
```

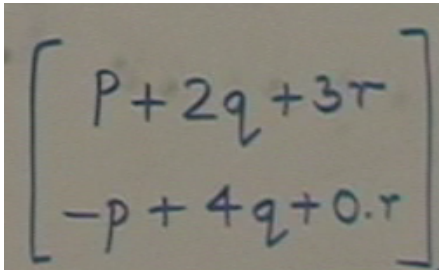
$$[\begin{array}{l}{P}\\ {q}\\ {r}\end{array}]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__rgb__product_vector

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT




```
\begin{bmatrix}p+2q+3r\\-p+4q+0\cdot r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix} p+2q+3r \\ -p+4q+0\cdot r \end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[P+2q+37]{-p49}+4q+0.7
```

$$[P+2q+37]-p49+4q+0.7$$

PP-FormulaNet_plus-S

Strict EM: False | Render: failed | Matrix cells: False

```
[left[P+2q+3.7\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[19+2q+3r]
```

$$[19+2q+3r]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[P+2q+3r]
```

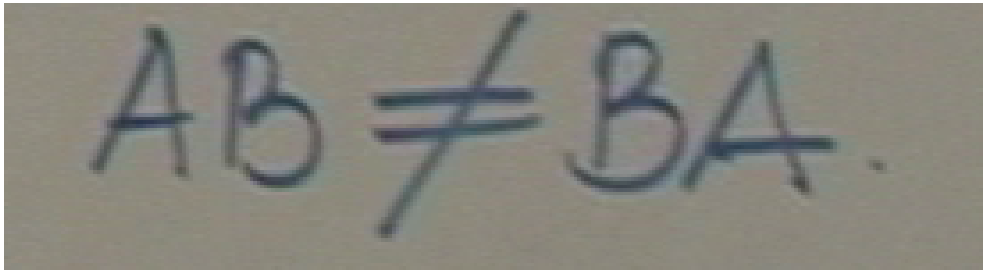
$$[P+2q+3r]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__rgb__noncommuting_product

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
AB\ne BA.
```

REFERENCE RENDERING

$$AB \neq BA.$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\ B A\cdot
```

$$AB\ BA\cdot$$

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: None

```
A B\because B A
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\neq B A.
```

$$AB \neq BA.$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\neq B A
```

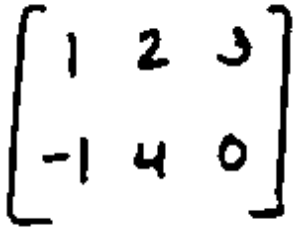
$$AB \neq BA$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__b0__rectangular_matrix

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}1&2&3\\-1&4&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\-1&4&0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[123]
```

$$[123]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
[int\limits_{[1Object{]}}^{[]2}]^{T}}\\ {\ 1\ 4H}\\ \end{aligned}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[123]
```

$$[123]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[-140]
```

$$[-140]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_07_67415__b0__rectangular_matrix

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\-1&4&0\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {-1}&{4}&{0}\\ \end{matrix}\right]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[{\begin{array}{l l l}{1}&{2}&{3}\\ {-1}&{4}&{0}\end{array}}\right]
```

REFERENCE RENDERING

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \end{bmatrix}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__b0__symbolic_vector

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\int_{i}^{p}$$

```
\int\limits_{i}^{p}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\int^{g} intfracg\frac{g}{z}$$

```
\int\limits_{}^{g}int frac{{g}}{{\frac{g}{z}}}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{q}]$$

```
[\frac{p}{q}]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{r}]$$

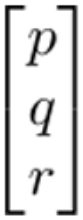
```
[\frac{p}{r}]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_07_67415__b0__symbolic_vector

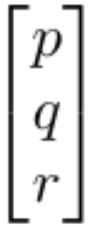
matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

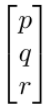
REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

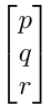
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

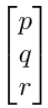
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

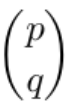
```
\left[\begin{matrix}{p}\\ {q}\\ {r}\\ \end{matrix}\right]
```



UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{p}{q}
```

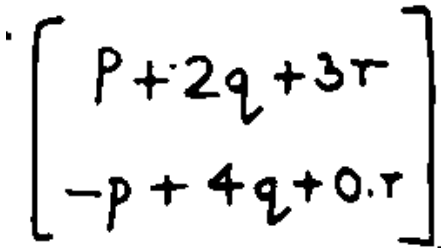


Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__b0__product_vector

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p+2q+3r\\-p+4q+0\cdot r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix} p+2q+3r \\ -p+4q+0\cdot r \end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\int\limits_{p+2q+3r}^{\ p+4q+0.r}^{\
```

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
[sqrt r geq2q+3\pi]^{-}]}{(-p+4q+0.r)}_{0}^{-1}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{p+2q+3r}{-p+4q+0.r}]
```

$$[\frac{p+2q+3r}{-p+4q+0.r}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[P+2q+3r]
```

$$[P+2q+3r]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_07_67415__b0__product_vector

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix} p+2q+3r \\ -p+4q+0\cdot r \end{bmatrix}$$

```
\begin{bmatrix}p+2q+3r\\-p+4q+0\cdot r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix} p+2q+3r \\ -p+4q+0\cdot r \end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{matrix}\right]
```

$$\left[\begin{matrix}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{array}{c}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{array}\right]
```

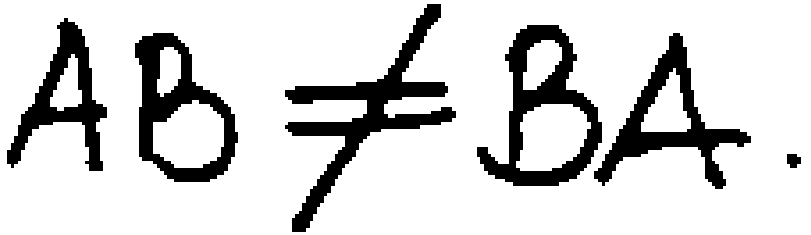
$$\left[\begin{array}{c}{p+2q+3r}\\ {-p+4q+0\cdot r}\end{array}\right]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_07_67415__b0__noncommuting_product

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$AB \neq BA.$$

```
AB\ne BA.
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$AB \neq \beta A.$$

```
A B\neq\beta A.
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$AB \neq BA.$$

```
A B\neq B A.
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$AB \neq BA.$$

```
A B\neq B A.
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$AB \neq BA.$$

```
A B\neq B A.
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_07_67415__b0__noncommuting_product

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



```
AB\ne BA.
```

REFERENCE RENDERING

$$AB \neq BA.$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\neq B A.
```

$$AB \neq BA.$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\neq B A.
```

$$AB \neq BA.$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\neq B A.
```

$$AB \neq BA.$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
A B\neq B A.
```

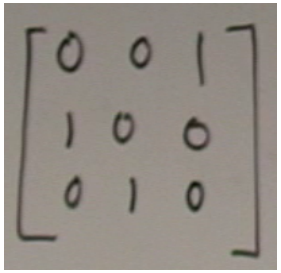
$$AB \neq BA.$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__rgb__permutation_matrix

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\begin{array}{l}{000}\\ {000}\\ {000}\\ \end{array}
```

$$\begin{array}{l}000\\000\\000\end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: failed | Matrix cells: False

```
[left0001\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{matrix}{0001}\\ {1000}\\ {010}\\ \end{matrix}]
```

$$[\begin{matrix}0001\\1000\\010\end{matrix}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[100]
```

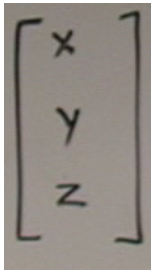
$$[100]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__rgb__input_vector

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}x\\y\\z\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}x\\y\\z\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\left[\begin{matrix}{x}\\ {y}\\ {z2}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{x}\\ {y}\\ {z2}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}]
```

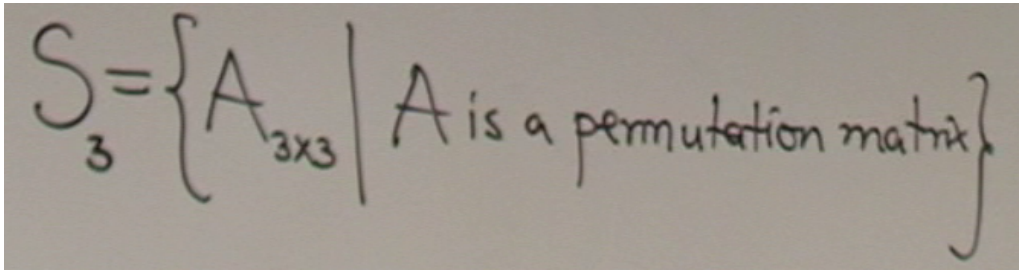
$$[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__rgb__set_builder

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\left\{A_{3\times3}\mid A\text{ is a permutation matrix}\right\}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$S_3 = \{A_{3\times3} | A \sin pn4 \leftrightarrow a \ \ \}^2$$

```
S_{3}=\{A_{3\times3}\vert A\sin{p n}4\ {\leftrightarrow}a{\ \ }{\ \\ \}}^{2}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$S_3 = \{A_{3\times3}\} A i s a \rho a m u u + \alpha n a \ + a a$$

```
S_{3}=\{A_{3\times3}\}A i s a\rho a m u u+\alpha n a\ +a a\\
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$S_3 = \{A_{3\times3} | A i s a p t m u t a t i o n \max\}$$

```
S_{3}=\{A_{3\times3}\vert A i s a p t m u t a t i o n\max\}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$S_3 = \{A_{3\times3} | A i s a p e m u l a d i o n m a t i o n\}$$

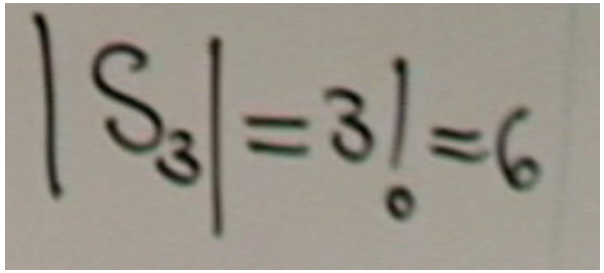
```
S_{3}=\{A_{3\times3}\vert A i s a p e m u l a d i o n m a t i o n\}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__rgb__cardinality_factorial

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert8_{3}\vert=31=6
```

$$|8_3| = 31 = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31vert=6
```

$$|S_3| = 31vert = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31=6
```

$$|S_3| = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=3\vert=6
```

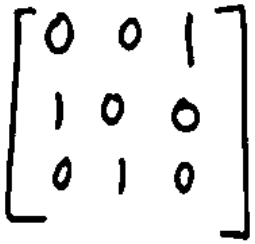
$$|S_3| = 3| = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__b0__permutation_matrix

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$$

```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\left.\begin{matrix}{0.01}\\ {0.00}\\ {0.0}\\ {0.0}\\ {end0}right\right
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[\begin{matrix}{001}\\ {1000}\\ {010}\\ \end{matrix}]$$

```
[\begin{matrix}{001}\\ {1000}\\ {010}\\ \end{matrix}]
```

UniMERNet

Strict EM: False | Render: failed | Matrix cells: False

```
\left[\begin{array}{l l}{0}&{0}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_08_53016__b0__permutation_matrix

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]
```

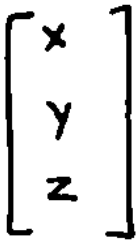
$$\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__b0__input_vector

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

```
\begin{bmatrix}x\\y\\z\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\left[\^{x}\right]_{z}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]$$

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}\right]$$

```
\left[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}\right]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

$$[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}]$$

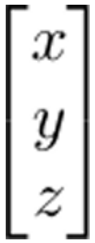
```
[\begin{array}{l}{x}\\ {y}\\ {z}\end{array}]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_08_53016__b0__input_vector

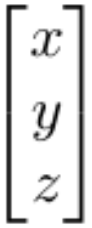
matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}x\\y\\z\end{bmatrix}
```

REFERENCE RENDERING



PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```



PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

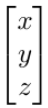
```
\left[\begin{matrix}{x}\\ {y}\\ {z}\\ \end{matrix}\right]
```



PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[{\begin{matrix}{x}\\ {y}\\ {z}\end{matrix}}\right]
```



UniMERNet

Strict EM: False | Render: unsupported | Matrix cells: False

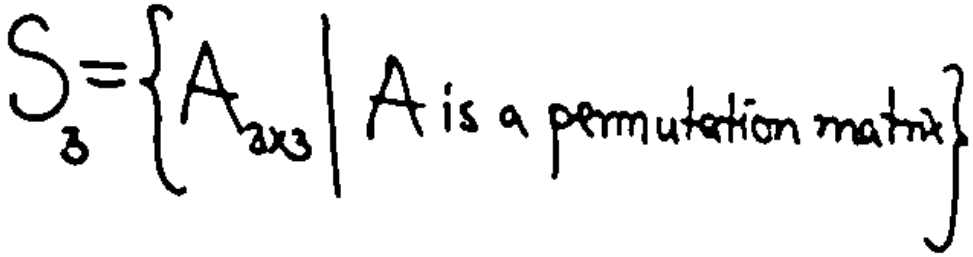
```
\left[{x\atop y}\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__b0__set_builder

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\left\{A_{3\times3}\mid A\text{ is a permutation matrix}\right\}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

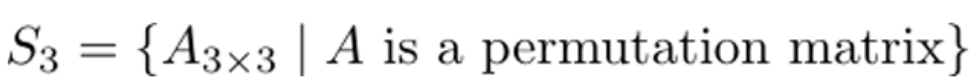
$$S_3 = \{ \lim_{A3 \to A \to B \sin} \ !\ !!\ !;!;$$

```
S_{3}=\{\lim\limits_{A3\rightarrow A\rightarrow B\sin\ \ {!!}}\ \ !\ !!\ !;!;\;\;\
```

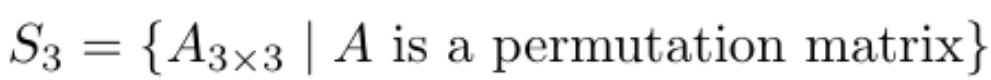
matched__lecture_08_53016__b0__set_builder

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT



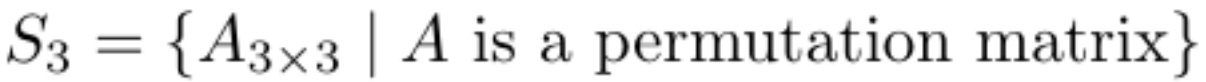
REFERENCE RENDERING



```
S_3=\left\{A_{3\times3}\mid A\text{ is a permutation matrix}\right\}
```

PP-FormulaNet-S

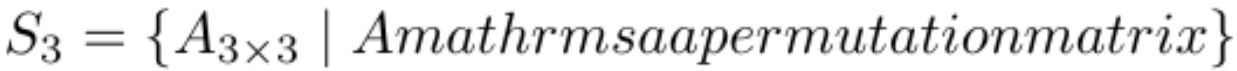
Strict EM: False | Render: rendered | Matrix cells: None



```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

PP-FormulaNet_plus-S

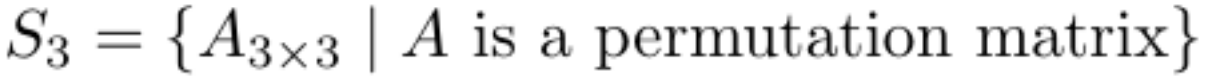
Strict EM: False | Render: rendered | Matrix cells: None



```
S_{3}=\{A_{3\times3}\mid A{mathrm s a a p e r m u t a t i o n m a t r i x}\}
```

PP-FormulaNet-L

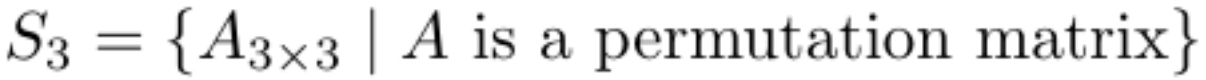
Strict EM: False | Render: rendered | Matrix cells: None



```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None



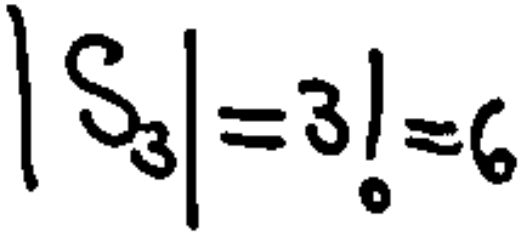
```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

real__lecture_08_53016__b0__cardinality_factorial

real_baseline | base | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: failed | Matrix cells: None

```
|vert__{3}\vert=3_{1}=6
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$|S_3| = 31 = G$$

```
|S_{3}|=31=G
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$|S_3| = 31 = 6$$

```
|S_{3}|=31=6
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$|s_3| = 3| = 6$$

```
\vert s_{3}\vert=3\vert=6
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

matched__lecture_08_53016__b0__cardinality_factorial

matched_print | matched_canvas_ink_bbox | no GT-based routing | all models, same input

ACTUAL INPUT

$$|S_3| = 3! = 6$$

```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

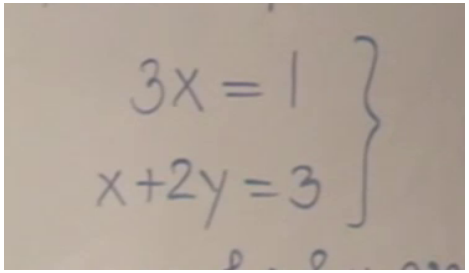
$$|S_3| = 3! = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_01_10423__rgb__formula_system__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT




```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3+2y=33$$

```
3+2y=33
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$x+2y=3$$

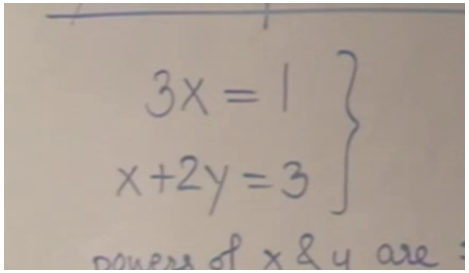
```
x+2y=3
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_01_10423__rgb__formula_system__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT




```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3==11$$

```
3==11
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=133$$

```
3x=133
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$x+2y=3\}$$

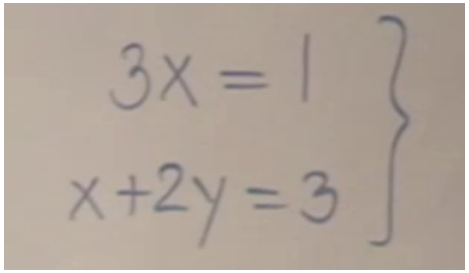
```
x+2y=3\}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_01_10423__rgb__formula_system__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT




```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3+2y=33$$

```
3+2y=33
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=13$$

```
3x=13
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$x+2y=3$$

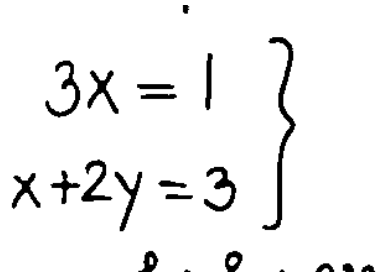
```
x+2y=3
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_01_10423__b0__formula_system__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\left.\begin{aligned}3x &= 1 \\ x+2y &= 3\end{aligned}\right\}$$

```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

PP-FormulaNet-S

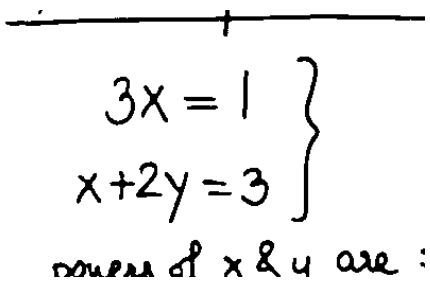
Strict EM: False | Render: unsupported | Matrix cells: None

```
3x=1\ \}\.,..\,\,\,
```

sensitivity__lecture_01_10423__b0__formula_system__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT




```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: None

```
\frac{3x=1}{x+2y=3}\ \ {\tan\\sin alpha
```

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: None

```
\frac{1}{3x+1sum yintint}^{}^{{x}}int^^{y}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\frac{1}{3x=1}\}x+2y=3\}
```

$$\frac{1}{3x=1}\}x+2y=3\}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\frac{3x-1}{x+2y=3}\}
```

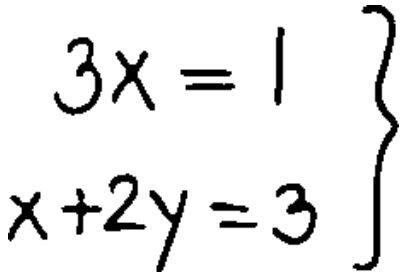
$$\frac{3x-1}{x+2y=3}\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_01_10423__b0__formula_system__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left.\begin{aligned}3x&=1\\x+2y&=3\end{aligned}\right\}
```

REFERENCE RENDERING

$$\left.\begin{aligned}3x &= 1\\ x+2y &= 3\end{aligned}\right\}$$

PP-FormulaNet-S

Strict EM: False | Render: failed | Matrix cells: None

```
3left+x=1\right\
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$3x=3$$

```
3x=3
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$\{3x=1\}\{x+2y=3\}$$

```
\{3x=1\}\{x+2y=3\}
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$x+2y=3$$

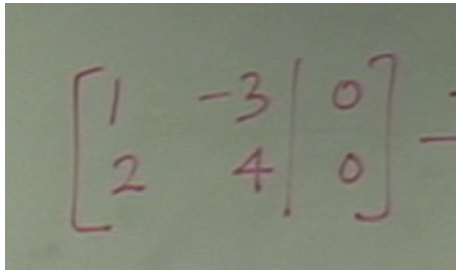
```
x+2y=3
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_06_47799__rgb__augmented_matrix__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2\ -\ 3110]]
```

$$[2\ -\ 3110]]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2-3]07
```

$$[2-3]07$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{matrix}{1}&{-3}\\ {2}&{4}\\ \end{matrix}]=
```

$$[\begin{matrix}{1}&{-3}\\ {2}&{4}\\ \end{matrix}]=$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[1,-3,0]=-24
```

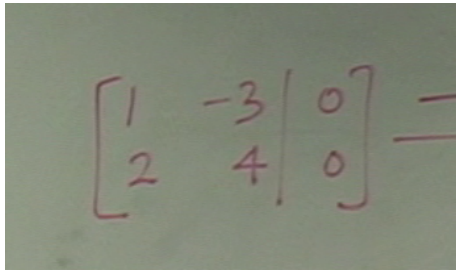
$$[1,-3,0]=-24$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_06_47799__rgb__augmented_matrix__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[240]
```

$$[240]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
22-307-
```

$$22 - 307 -$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{1}{2}-3\vert0]=
```

$$[\frac{1}{2} - 3|0] =$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[1,-3,10]=
```

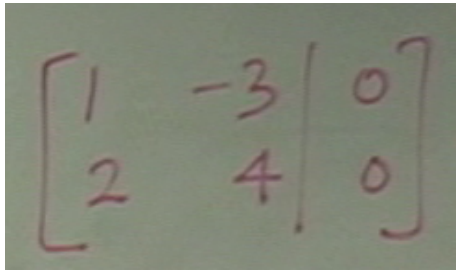
$$[1, -3, 10] =$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_06_47799__rgb__augmented_matrix__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2-310]
```

$$[2-310]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2-3]07
```

$$[2-3]07$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[1-3\vert0]
```

$$[1-3|0]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[1,-3,0]
```

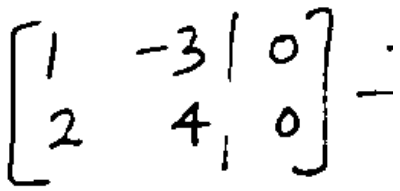
$$[1,-3,0]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_06_47799__b0__augmented_matrix__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
[2\ \ frac\vert-3\ 0]{
```

PP-FormulaNet_plus-S

Strict EM: False | Render: failed | Matrix cells: False

```
[2^{}frac-3]^{o}]^
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[\begin{matrix}1&-3\\2&4\end{matrix}]-$$

```
[\begin{matrix}{1}&{-3}\\ {2}&{4}\\ \end{matrix}]-
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$[1,-3,10]-$$

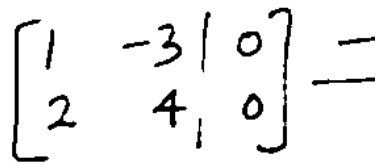
```
[1,-3,10]-
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_06_47799__b0__augmented_matrix__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[\sum{2}\frac{-3}{4}]=
```

$$[\sum 2\frac{-3}{4}] =$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2^{hat x_,{j}},^{\hat{y}_{j}}]^{\hat{x}_{j}}
```

$$[2^{hatx_,j},^{\hat{y}_j}]^{\hat{x}_j}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[1-3]0]=
```

$$[1-3]0] =$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[1,-3,10]=
```

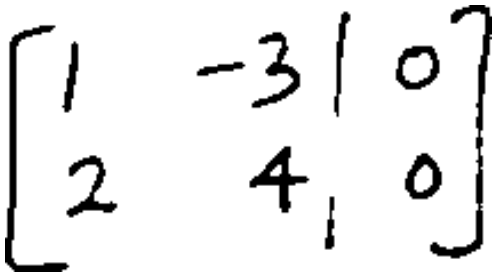
$$[1,-3,10] =$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_06_47799__b0__augmented_matrix__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\left[\begin{array}{cc|c}1&-3&0\\2&4&0\end{array}\right]
```

REFERENCE RENDERING

$$\left[\begin{array}{cc|c} 1 & -3 & 0 \\ 2 & 4 & 0 \end{array}\right]$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2,310]]
```

$$[2, 310]]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[2,\ frac-3]0o]
```

$$[2,\ frac - 3]0o]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[1-310]
```

$$[1 - 310]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{array}{l l l}{1}&{-3\vert}&{0}\\ {2}&{4}&{0}\end{array}]
```

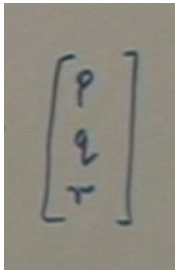
$$\left[\begin{array}{lll} 1 & -3\vert & 0 \\ 2 & 4 & 0 \end{array}\right]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__rgb__symbolic_vector__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$[frac99]$$

```
\ [frac{9}{9}]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$1\ 9$$

```
\ 1\ 9
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{q}]$$

```
[\frac{p}{q}]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$[\rho]$$

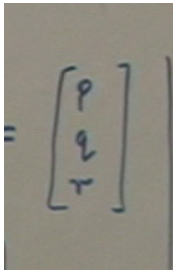
```
[\rho]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__rgb__symbolic_vector__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$= 971$$

```
=971
```

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
F=\left[\begin{matrix}{P}\\ {end matrix}]\\right
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$= [\frac{p}{q}]$$

```
=[\frac{p}{q}]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$= [\frac{p}{q}]$$

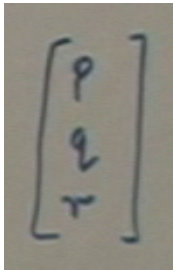
```
=[\frac{p}{q}]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__rgb__symbolic_vector__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\left[\begin{matrix}{9}\\ {9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{9}\\ {9}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
l
```

$$l$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{9}{9}]
```

$$[\frac{9}{9}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{array}{l}{P}\\ {q}\\ {r}\end{array}]
```

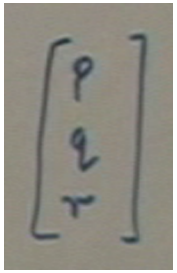
$$[\begin{array}{l}{P}\\ {q}\\ {r}\end{array}]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__rgb__symbolic_vector__scale2

sensitivity | scale2 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\left[\begin{matrix}{9}\\ {9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{9}\\ {9}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
l7
```

$$l7$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{9}{9}]
```

$$[\frac{9}{9}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{array}{l}{P}\\ {q}\\ {r}\end{array}]
```

$$[\begin{array}{l}{P}\\ {q}\\ {r}\end{array}]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__b0__symbolic_vector__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\int_{i}^{p}$$

```
\int\limits_{i}^{p}
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\int^{g} intfracg\frac{g}{z}$$

```
\int\limits_{}^{g}int frac{{g}}{{\frac{g}{z}}}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{q}]$$

```
[\frac{p}{q}]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{r}]$$

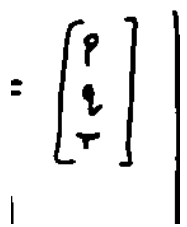
```
[\frac{p}{r}]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__b0__symbolic_vector__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
f[i\]^{j}]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
|int sum int|||||
```

$$|intsumint|||||$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
=[\frac{p}{q}]
```

$$=[\frac{p}{q}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
=\binom{p}{r}
```

$$=\binom{p}{r}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__b0__symbolic_vector__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}p\\q\\r\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\int\limits_{i}^{p}
```

$$\int_{i}^{p}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\int\limits_{}^{g}int frac{{g}}{{\frac{g}{z}}}
```

$$\int_{}^{g} int frac \frac{g}{\frac{g}{z}}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{p}{q}]
```

$$[\frac{p}{q}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[\frac{p}{r}]
```

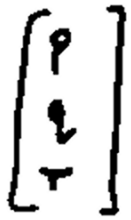
$$[\frac{p}{r}]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_07_67415__b0__symbolic_vector__scale2

sensitivity | scale2 | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

```
\begin{bmatrix}p\\q\\r\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\int [limits_{i}^{p}]$$

```
\int[limits_{i}^{p}]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$\int\limits_{}^{g} intfracgL[$$

```
\int\limits_{}^{g}int frac{g}{L}[
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{q}]$$

```
[\frac{p}{q}]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$[\frac{p}{r}]$$

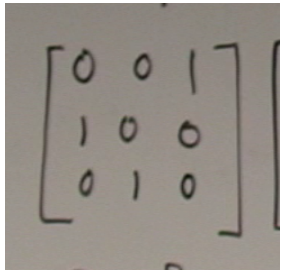
```
[\frac{p}{r}]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__permutation_matrix__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[0.00]]{}
```

$$[0.00]]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
[0011]
```

$$[0011]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{matrix}{0}&{0}\\ {1}&{0}\\ {0}&{1}&{0}\\ \end{matrix}]
```

$$\begin{matrix}0 & 0 & \\ [1 & 0 &]\\ 0 & 1 & 0\end{matrix}$$

UniMERNet

Strict EM: False | Render: failed | Matrix cells: True

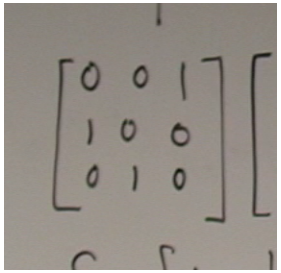
```
[\begin{array}{l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__permutation_matrix__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

$$[0.01][$$

```
[0.01][
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

$$[frac1000][0][$$

```
[frac{1}{0}00][0][
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$\begin{matrix}0&1\\1&0\\0&1\\0&1\end{matrix}[$$

```
[\begin{matrix}{0}&{1}\\ {1}&{0}\\ {0}&{1}&{0}\\ {0}&{1}&{1}\\ \end{matrix}][
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

$$\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]\left[\begin{array}{l l l}{\frac{1}{r}}&{\frac{1}{\theta}}&{\frac{1}{r}}\\ {\frac{1}{\theta}}&{0}&{\frac{1}{\phi}}\\ {0}&{1}&{0}\end{array}\right]$$

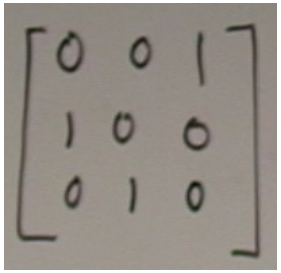
```
\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]\left[\begin{array}{l l
l}{\frac{1}{r}}&{\frac{1}{\theta}}&{\frac{1}{r}}\\ {\frac{1}{\theta}}&{0}&{\frac{1}{\phi}}\\ {0}&{1}&{0}\end{array}\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__permutation_matrix__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\begin{array}{l}{000}\\ {000}\\ {000}\\ \end{array}
```

$$\begin{array}{l}{000}\\ {000}\\ {000}\\ \end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: failed | Matrix cells: False

```
[left0001\right]
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{matrix}{0001}\\ {1000}\\ {010}\\ \end{matrix}]
```

$$[\begin{matrix}{0001}\\ {1000}\\ {010}\\ \end{matrix}]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
[100]
```

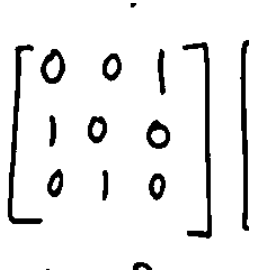
$$[100]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__permutation_matrix__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: failed | Matrix cells: False

```
\left.\ \ {\begin{matrix}{0017}\\ {010}\\ {0}\\ {01\\ \end{matrix}}}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: failed | Matrix cells: False

```
\left[\begin{matrix}{0}&{0}&{0}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ {}&{0&{0}}\\ \end{matrix}\right]_
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$\begin{matrix}0 & 0\\ [1 & 0]\\ 0 & 1\end{matrix}$$

```
[\begin{matrix}{0}&{0}\\ {1}&{0}\\ {0}&{1}\\ \end{matrix}]
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

$$\left[\begin{array}{l l l}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{array}\right]1$$

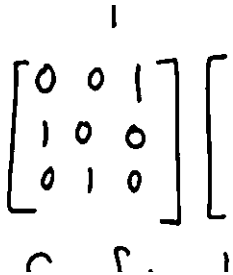
```
\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]1
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__permutation_matrix__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$\begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$$

```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\left\lbrack\begin{matrix}{1001}\\ {0010}\\ {c0}}}\\{\}}\\end{}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\begin aligned}{{}&{{}\ leftlceil[\beginbeginmatrix}}{{{{0}}&{{{{0}}}}&{{{{1}}}}\\ {{{{}1}}}&{{{{0}}}}&{{{{0}}}}\\ {{{{}0}}}&{{{{1}}}}&{{{{0}}}}\\
\end{matrix}\right]_{mathrm}^{2}}\\ {}&{{}=}}\\ \end{aligned}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

$$\begin{matrix}0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 1 & 0\end{matrix}\left[\right.\left[\right.$$

```
[\begin{matrix}{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\\ {0}&{1}&{0}\\ \end{matrix}][
```

UniMERNet

Strict EM: False | Render: failed | Matrix cells: False

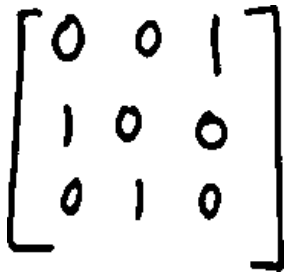
```
|\limits_{\left[\begin{array}{l l l}{0}&{0}&{1}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]}|\limits_{\left[\begin{array}{l l
l}{\frac{1}{r}}&{\frac{1}{2}}&{0}\\ {0}&{1}&{0}\\ {\frac{1}{3}}&{0}&{1}\end{array}\right]}
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__permutation_matrix__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: unsupported | Matrix cells: False

```
\left.\begin{matrix}{0.01}\\ {0.00}\\ {0.0}\\ {0.0}\\ {end0}right\right
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{0}&{1}\\ {0}&{0}\\ {0}&{0}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
[\begin{matrix}{001}\\ {1000}\\ {010}\\ \end{matrix}]
```

$$[\begin{matrix}{001}\\ {1000}\\ {010}\\ \end{matrix}]$$

UniMERNet

Strict EM: False | Render: failed | Matrix cells: False

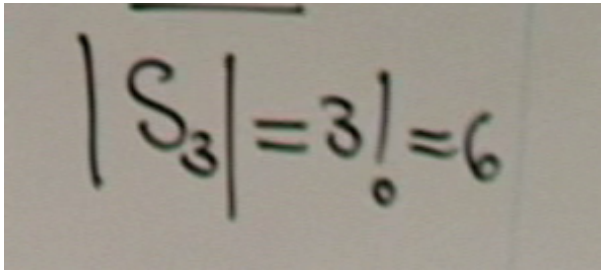
```
\left[\begin{array}{l l}{0}&{0}\\ {1}&{0}&{0}\\ {0}&{1}&{0}\end{array}\right]
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__cardinality_factorial__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert\overline{S_{3}}\vert=31=6
```

$$|\overline{S_3}| = 31 = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31vert=6
```

$$|S_3| = 31vert = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31=6
```

$$|S_3| = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=3\vert=6
```

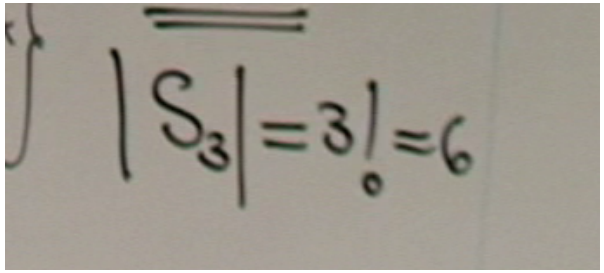
$$|S_3| = 3| = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__cardinality_factorial__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert\overrightarrow{s_{3}\vert=3}\vert=6
```

$$\overrightarrow{|s_3| = 3}| = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\ \vert s_{3}\vert=31=6
```

$$|s_3| = 31 = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\sqrt{S_{3}1}=31=6
```

$$\sqrt{S_3 1} = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\int\limits_{0}^{\infty}\vert\overline{{S_{3}}}\vert=31=6
```

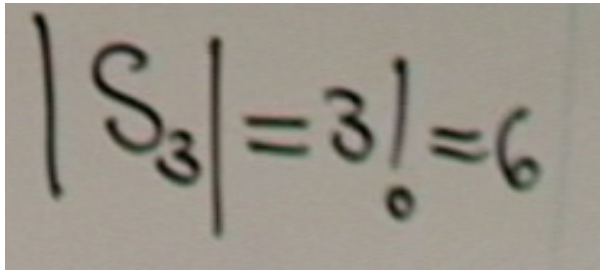
$$\int_{0}^{\infty} |\overline{S_3}| = 31 = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__cardinality_factorial__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert8_{3}\vert=31=6
```

$$|8_3| = 31 = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31vert=6
```

$$|S_3| = 31vert = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31=6
```

$$|S_3| = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=3\vert=6
```

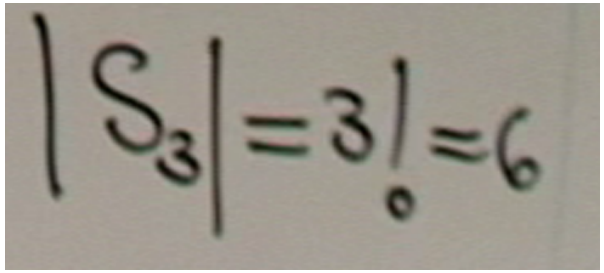
$$|S_3| = 3| = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__rgb__cardinality_factorial__scale2

sensitivity | scale2 | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert8_{3}\vert=31=6
```

$$|8_3| = 31 = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31vert=6
```

$$|S_3| = 31vert = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=31=6
```

$$|S_3| = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert S_{3}\vert=3\vert=6
```

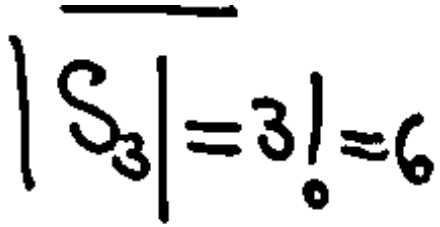
$$|S_3| = 3| = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__cardinality_factorial__pad10

sensitivity | pad10 | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$|S_3| = 3! = 6$$

```
|S_3|=3!=6
```

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

$$|vert_3 3vert = 31 = 6$$

```
|vert_33vert=31=6
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$|s_3| = 31 = 6$$

```
|s_{3}|=31=6
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$|S_3| = 31 = 6$$

```
\vert S_{3}\vert=31=6
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$|s_3| = 31 \approx 6$$

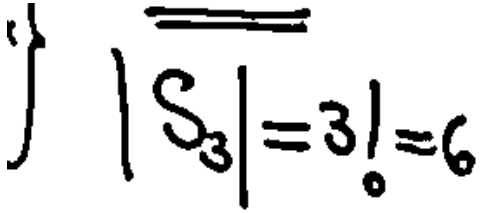
```
\vert s_{3}\vert=31\approx6
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__cardinality_factorial__pad20

sensitivity | pad20 | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert\vert\overline{s_{3}}\vert=3\vert=6
```

$$||\overline{s_3}| = 3| = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
dint|s_{3}|=31=0
```

$$dint|s_3| = 31 = 0$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\sqrt[3]{|S_{3}|}=31=6
```

$$\sqrt[3]{|S_3|} = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\int\limits_{0}^{1}|\overline{S_{3}}|=31=6
```

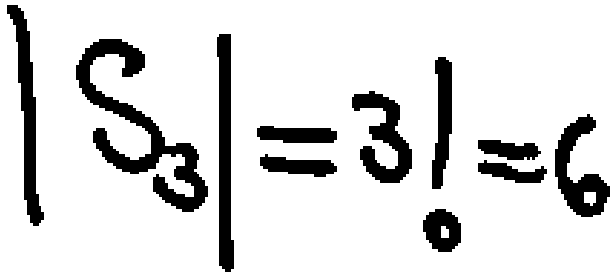
$$\int_0^1 |\overline{S_3}| = 31 = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__cardinality_factorial__tight

sensitivity | tight | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: failed | Matrix cells: None

```
|vert__{3}\vert=3_{1}=6
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

$$|S_3| = 31 = G$$

```
|S_{3}|=31=G
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

$$|S_3| = 31 = 6$$

```
|S_{3}|=31=6
```

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

$$|s_3| = 3| = 6$$

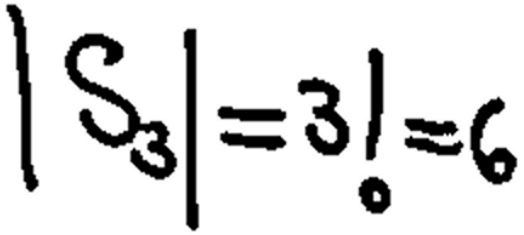
```
\vert s_{3}\vert=3\vert=6
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

sensitivity__lecture_08_53016__b0__cardinality_factorial__scale2

sensitivity | scale2 | no GT-based routing | all models, same input

ACTUAL INPUT



```
|S_3|=3!=6
```

REFERENCE RENDERING

$$|S_3| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|vert\ _{3}\vert=3\vert_{0}=6
```

$$|vert\ _3| = 3|_0 = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=31=G
```

$$|S_3| = 31 = G$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=31=6
```

$$|S_3| = 31 = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\vert s_{3}\vert=3\vert=6
```

$$|s_3| = 3| = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__vec3__wide_canvas

print_format | extra_blank_canvas | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1\\2\\3\end{bmatrix}$$

```
\begin{bmatrix}1\\2\\3\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1\\2\\3\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}\\ {2}\\ {3}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\binom{1}{2}
```

$$\binom{1}{2}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__matrix33__wide_canvas

print_format | extra_blank_canvas | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

```
\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\\ \end{matrix}\right]$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: True

```
\left[\begin{array}{l l l}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\end{array}\right]
```

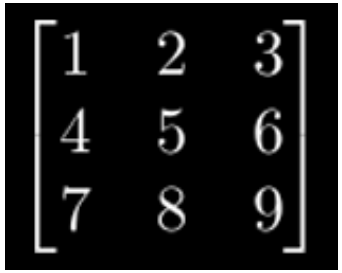
$$\left[\begin{array}{l l l}{1}&{2}&{3}\\ {4}&{5}&{6}\\ {7}&{8}&{9}\end{array}\right]$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__matrix33__inverted

print_format | inverted | no GT-based routing | all models, same input

ACTUAL INPUT



```
\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}
```

REFERENCE RENDERING

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: False

```
39
```

$$39$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: False

```
\frac{1}{2}\ \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}
```

$$\frac{1}{2}\ \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: False

```
\frac{1}{2}
```

$$\frac{1}{2}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: False

```
\angle A C B=\angle C B D
```

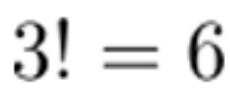
$$\angle ACB = \angle CBD$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__factorial__wide_canvas

print_format | extra_blank_canvas | no GT-based routing | all models, same input

ACTUAL INPUT



3!=6

REFERENCE RENDERING

$$3! = 6$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3! = 6$$

3!=6

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3! = 6$$

3!=6

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3! = 6$$

3!=6

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3! = 6$$

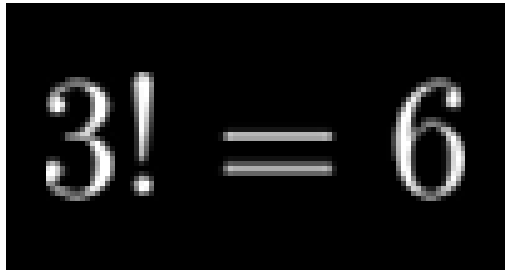
3!=6

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__factorial__inverted

print_format | inverted | no GT-based routing | all models, same input

ACTUAL INPUT



```
3!=6
```

REFERENCE RENDERING

$$3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
B square(c0square
```

$$Bsquare(c0square$$

PP-FormulaNet_plus-S

Strict EM: False | Render: unsupported | Matrix cells: None

```
8textcircled{8}\ circ{1}\ circ{1}\textcircled{6}
```

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
B\equiv(0)
```

$$B \equiv (0)$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
3x=0
```

$$3x = 0$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__indexed_factorial__wide_canvas

print_format | extra_blank_canvas | no GT-based routing | all models, same input

ACTUAL INPUT

$$|S_3| = 3! = 6$$

```
|S_3|=3!=6
```

REFERENCE RENDERING

$$\left|S_3\right| = 3! = 6$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
|S_{3}|=3!=6
```

$$|S_3| = 3! = 6$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__lines2__wide_canvas

print_format | extra_blank_canvas | no GT-based routing | all models, same input

ACTUAL INPUT

$$\begin{aligned}3x&=1\\x+2y&=3\end{aligned}$$

```
\begin{aligned}3x&=1\\x+2y&=3\end{aligned}
```

REFERENCE RENDERING

$$\begin{aligned}3x&=1\\x+2y&=3\end{aligned}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{{x}=1,}\\ {{x}+2{y}=3.}\\ \end{array}
```

$$\begin{array}{r}{{x}=1,}\\ {{x}+2{y}=3.}\\ \end{array}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}
```

$$\begin{array}{r}{3x=1}\\ {x+2y=3}\end{array}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

format__set_text__wide_canvas

print_format | extra_blank_canvas | no GT-based routing | all models, same input

ACTUAL INPUT

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

```
S_3=\{A_{3\times3}\mid A\text{ is a permutation matrix}\}
```

REFERENCE RENDERING

$$S_3 = \{A_{3\times3} \mid A \text{ is a permutation matrix}\}$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A\mathrm{~i\ s~a~p\ e\ r\ m\ u\ t\ a\ t\ i\ o\ n~m\ a\ t\ r\ i\ x}\}$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{mathrm s a a p e r m u t a t i o n m a t r i x}\}
```

$$S_3 = \{A_{3\times3} \mid A\mathrm{mathrm\ s\ a\ a\ p\ e\ r\ m\ u\ t\ a\ t\ i\ o\ n\ m\ a\ t\ r\ i\ x}\}$$

PP-FormulaNet-L

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A\mathrm{~i\ s~a~p\ e\ r\ m\ u\ t\ a\ t\ i\ o\ n~m\ a\ t\ r\ i\ x}\}$$

UniMERNet

Strict EM: False | Render: rendered | Matrix cells: None

```
S_{3}=\{A_{3\times3}\mid A{\mathrm{~i s~a~p e r m u t a t i o n~m a t r i x}}\}
```

$$S_3 = \{A_{3\times3} \mid A\mathrm{~i\ s~a~p\ e\ r\ m\ u\ t\ a\ t\ i\ o\ n~m\ a\ t\ r\ i\ x}\}$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__lines2__row0

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



3x=1

REFERENCE RENDERING

$$3x = 1$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

3x=1

$$3x = 1$$

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

3x=1

$$3x = 1$$

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

3x=1

$$3x = 1$$

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

3x=1

$$3x = 1$$

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__lines2__row1

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



x+2y=3

REFERENCE RENDERING

$$x + 2y = 3$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

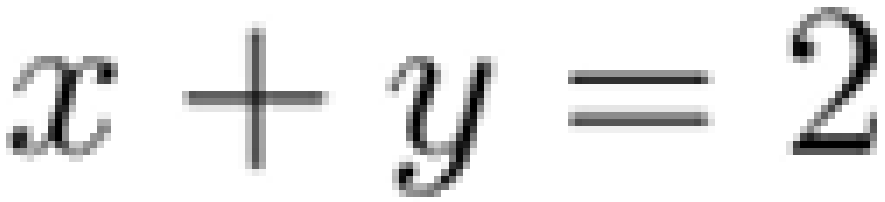
x+2y=3

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__lines3__row0

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



x+y=2

REFERENCE RENDERING

$$x + y = 2$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + y = 2$$

x+y=2

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + y = 2$$

x+y=2

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$x + y = 2$$

x+y=2

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$x + y = 2$$

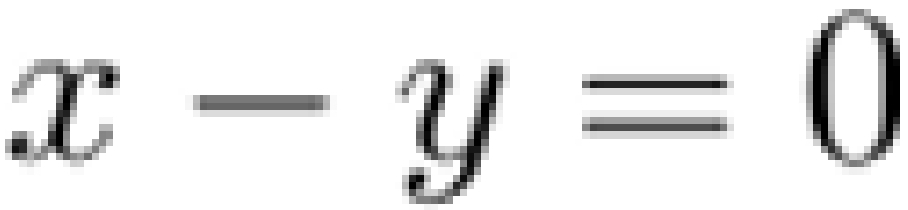
x+y=2

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__lines3__row1

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$x - y = 0$$

```
x-y=0
```

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x - y = 0$$

```
x-y=0
```

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x - y = 0$$

```
x-y=0
```

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$x - y = 0$$

```
x-y=0
```

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$x - y = 0$$

```
x-y=0
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__lines3__row2

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



z=3

REFERENCE RENDERING

$$z = 3$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$z = 3$$

z=3

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$z = 3$$

z=3

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$z = 3$$

z=3

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$z = 3$$

z=3

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__matrix22__row0

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



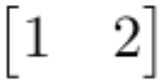
REFERENCE RENDERING



```
1\quad2
```

PP-FormulaNet-S

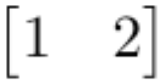
Strict EM: False | Render: rendered | Matrix cells: None



```
\left[\begin{matrix}{1}&{2}\\ \end{matrix}\right]
```

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None



```
\left[\begin{matrix}{1}&{2}\\ \end{matrix}\right]
```

PP-FormulaNet-L

Strict EM: False | Render: unsupported | Matrix cells: None

```
\left\lceil\!\!1\quad2\!\!\right\rceil
```

UniMERNet

Strict EM: False | Render: unsupported | Matrix cells: None

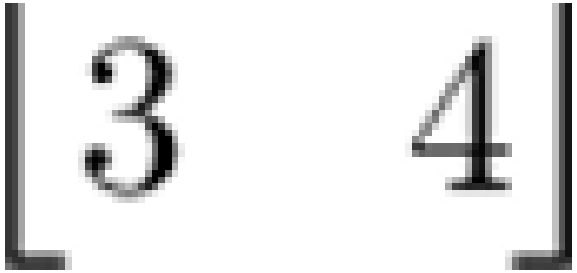
```
\left\lceil1\quad2\right\rceil
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__matrix22__row1

split_row | diagnostic_fixed_equal_row_crop | no GT-based routing | all models, same input

ACTUAL INPUT



```
3\quad4
```

REFERENCE RENDERING

$$3 \quad 4$$

PP-FormulaNet-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\left[\begin{matrix}{3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{3}&{4}\\ \end{matrix}\right]$$

PP-FormulaNet_plus-S

Strict EM: False | Render: rendered | Matrix cells: None

```
\left[\begin{matrix}{3}&{4}\\ \end{matrix}\right]
```

$$\left[\begin{matrix}{3}&{4}\\ \end{matrix}\right]$$

PP-FormulaNet-L

Strict EM: False | Render: unsupported | Matrix cells: None

```
\left\lfloor3\quad4\right\rfloor
```

UniMERNet

Strict EM: False | Render: unsupported | Matrix cells: None

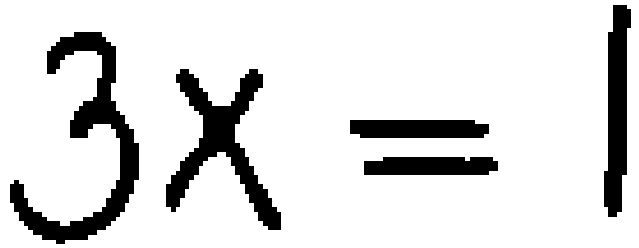
```
\lfloor3\ \ \ 4\rfloor
```

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__hand_system__row0

split_row | v25_geometric_split_retest | no GT-based routing | all models, same input

ACTUAL INPUT



3x=1

REFERENCE RENDERING

$$3x = 1$$

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

3x=1

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

3x=1

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

3x=1

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$3x = 1$$

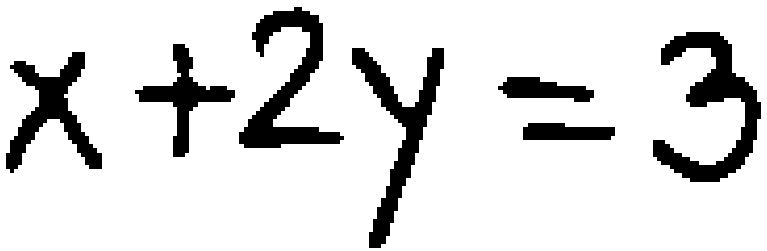
3x=1

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.

split__hand_system__row1

split_row | v25_geometric_split_retest | no GT-based routing | all models, same input

ACTUAL INPUT



REFERENCE RENDERING

$$x + 2y = 3$$

x+2y=3

PP-FormulaNet-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

PP-FormulaNet_plus-S

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

PP-FormulaNet-L

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

UniMERNet

Strict EM: True | Render: rendered | Matrix cells: None

$$x + 2y = 3$$

x+2y=3

Render success / matrix cell agreement / raster similarity are separate observations, not mathematical correctness verdicts.